

Multiple Goal Analysis Using Robot Heading

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Abstract

This project extends the single-goal geometric analysis introduced in Project 1 to an environment containing multiple candidate goals. A robot heading vector is introduced to represent the direction in which the robot is facing. For each candidate goal, the scalar product between the robot heading and the normalized direction toward the goal is computed. This value is interpreted as the cosine of the angle between both unit vectors and is used to classify each goal as being in front of, beside, or behind the robot.

1 Introduction

A mobile robot may operate in an environment containing several possible target positions. In this situation, selecting a goal cannot depend only on the distance between the robot and each candidate. The robot must also consider its current orientation, because a nearby goal located behind the robot may be less relevant than a goal positioned within its forward field.

This project introduces a robot heading vector and uses it to evaluate the relative position of multiple candidate goals. Each goal is processed individually, classified according to its position with respect to the robot heading, and tested using a visibility condition.

The geometric operations developed in Project 1 are reused to obtain the distance and normalized direction associated with each candidate goal. The new contribution of this project is therefore the combination of robot orientation, scalar-product analysis, spatial classification and visibility filtering.

2 Theoretical Background

2.1 Multiple Goal Processing

Instead of considering only one target, the robot evaluates a set of candidate goal positions:

$$\mathbf{p}G_1, \mathbf{p}G_2, \dots, \mathbf{p}G_n$$

Each candidate is processed independently. For goal i , the previously introduced geometric operations provide

the normalized direction vector from the robot toward that goal:

$$\hat{\mathbf{v}}_{RG_i}$$

2.2 Robot Heading

The robot heading is represented by the vector

$$\mathbf{h} = \begin{bmatrix} h_x \\ h_y \end{bmatrix}$$

The heading is a unit vector:

$$\|\mathbf{h}\| = 1$$

Introducing orientation is necessary because the position of a goal alone does not indicate whether that goal lies in front of or behind the robot. Two goals may be equally distant but located in completely different directions relative to the robot heading.

2.3 Scalar Product

For each candidate goal i , the scalar product between the robot heading and the normalized goal direction is calculated as

$$s_i = \mathbf{h} \cdot \hat{\mathbf{v}}_{RG_i}$$

Because both vectors are unit vectors, the scalar product can be interpreted as

$$s_i = \cos(\alpha_i)$$

where α_i is the angle between the robot heading and the direction toward goal i .

$$-1 \leq s_i \leq 1$$

This value therefore provides a compact numerical description of the angular relationship between the robot orientation and each goal direction.

2.4 Geometric Interpretation of the Scalar Product

The scalar-product value has a direct geometric interpretation. When $s_i = 1$, the goal lies directly in front of the robot and is aligned with the robot heading. When $s_i = 0$, the goal direction is perpendicular to the robot heading. When $s_i = -1$, the goal lies directly behind the robot. Positive scalar-product values indicate that the goal is located in the forward half-plane of the robot. Negative values indicate that the goal lies in the rear half-plane. Values close to one describe strong alignment with the robot heading, while values close to zero describe a more lateral goal position.

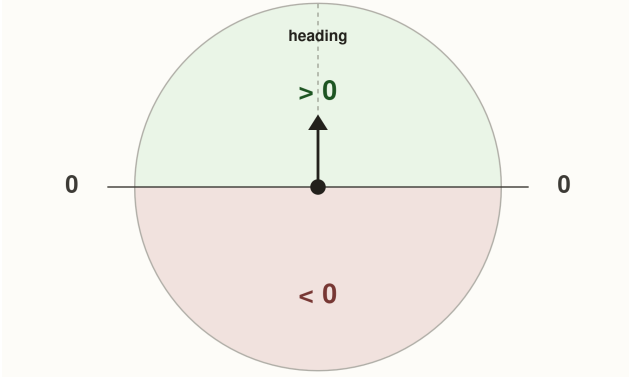


Figure 1: Forward-Backward

2.5 Spatial Classification and Goal Visibility

The experiment applies the following spatial classification rules:

$$0 \leq s_i \leq 0.707 \Rightarrow \text{goal beside the robot,} \quad (1)$$

$$s_i > 0.707 \Rightarrow \text{goal in front of the robot,} \quad (2)$$

$$s_i < 0 \Rightarrow \text{goal behind the robot.} \quad (3)$$

The threshold 0.707 is a design choice corresponding approximately to an angular separation of 45° , since $\cos(45^\circ) \approx 0.707$. It separates strongly forward-facing goals from goals that are still located in the forward half-plane but are positioned more laterally.

Spatial classification and visibility are related but distinct. A goal is considered visible by the selection algorithm only when

$$s_i \geq 0 \quad (4)$$

and

$$d_i > 0. \quad (5)$$

The first condition restricts selection to goals in the forward half-plane. The second condition prevents the robot position itself from being selected as a valid goal.

This distinction is important. A goal may be classified as beside the robot because its scalar product is zero, yet still be rejected when its distance is also zero.

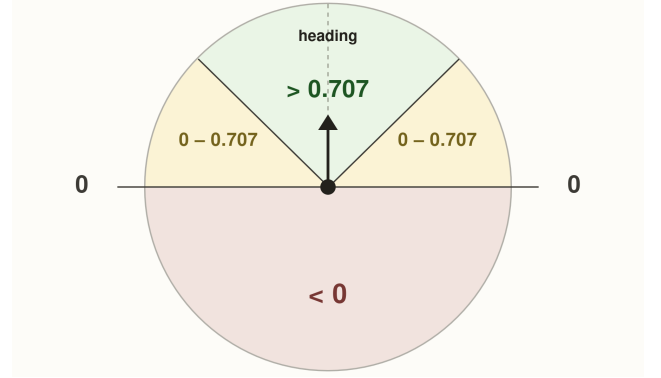


Figure 2: Forward-Backward-Beside

3 Project Task

Define one robot position, a unit heading vector, and several candidate goal positions as NumPy arrays. Process all goals iteratively, reuse the geometric calculations from Project 1 to obtain each distance and normalized goal direction, and calculate the scalar product between the robot heading and every goal direction. Classify each goal as in front of, beside, or behind the robot, filter the goals using the visibility conditions, and select the closest visible goal. Finally, print the calculated values, classifications, and selected target.

References

- [1] J. Quejido Barrera, *Vector-Based Goal Representation for a Mobile Robot*, Road to become more, Project 1, unpublished technical report, 2026.